

33rd Nordic Network Meeting on “String, Fields and Branes”

Mass gap puzzle for a near extremal Lifshitz black hole

Matthias Harsen

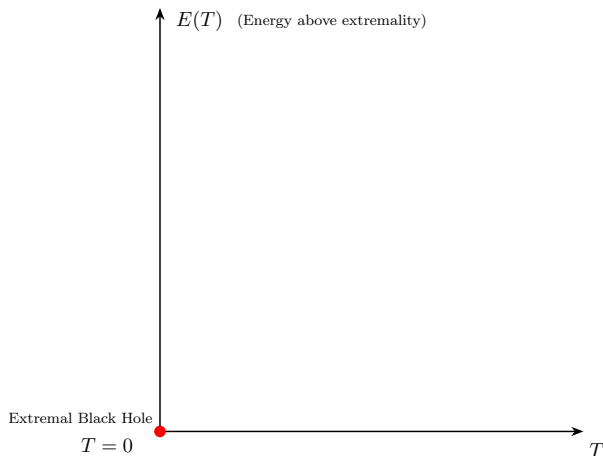
Based on [\[2408.15336\]](#) with Watsé Sybesma

Mass gap puzzle for Reissner–Nordström Black Holes



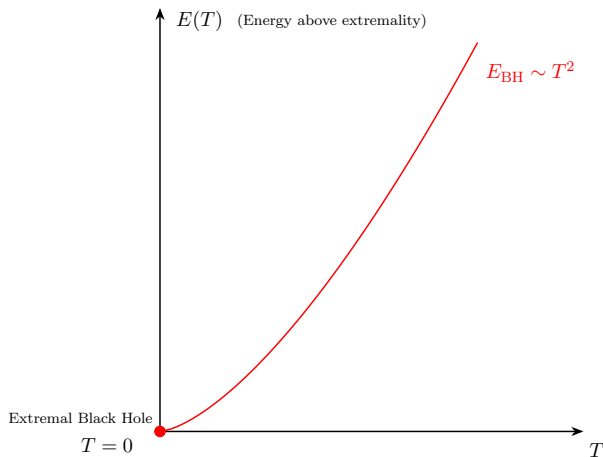
[Preskill, Schwarz, Shapere, Trivedi, Wilczek; 91]

Mass gap puzzle for Reissner–Nordström Black Holes



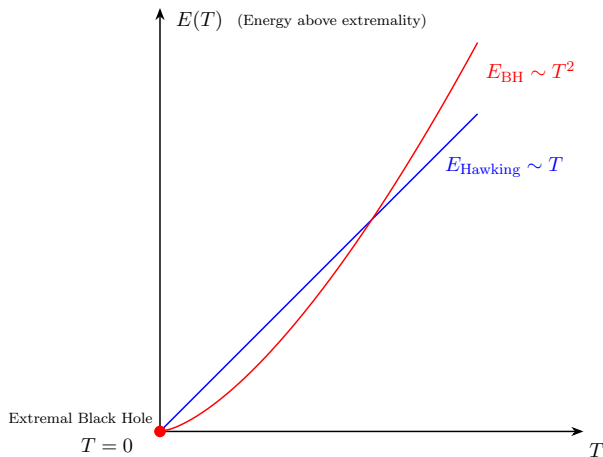
[Preskill, Schwarz, Shapere, Trivedi, Wilczek; 91]

Mass gap puzzle for Reissner–Nordström Black Holes



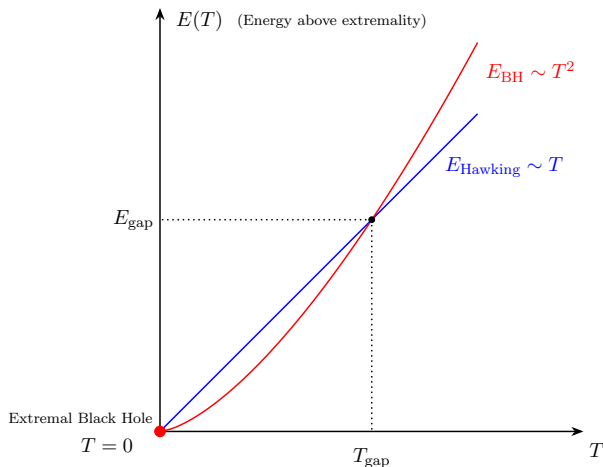
[Preskill, Schwarz, Shapere, Trivedi, Wilczek; 91]

Mass gap puzzle for Reissner–Nordström Black Holes



[Preskill, Schwarz, Shapere, Trivedi, Wilczek; 91]

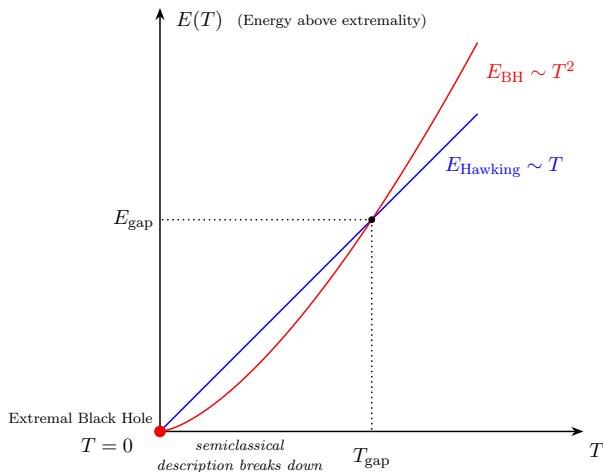
Mass gap puzzle for Reissner–Nordström Black Holes



[Preskill, Schwarz, Shapere, Trivedi, Wilczek; 91]



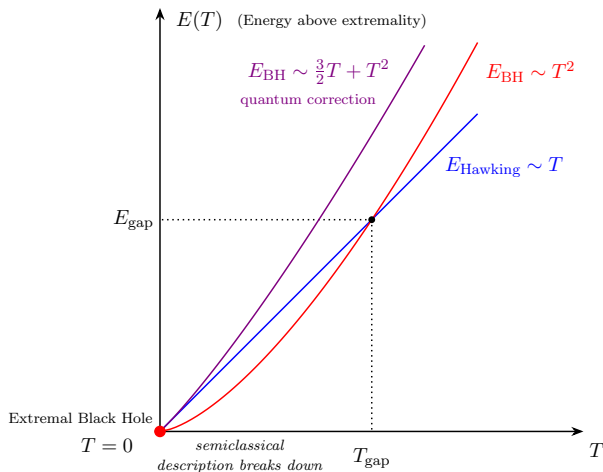
Mass gap puzzle for Reissner–Nordström Black Holes



[Preskill, Schwarz, Shapere, Trivedi, Wilczek; 91]



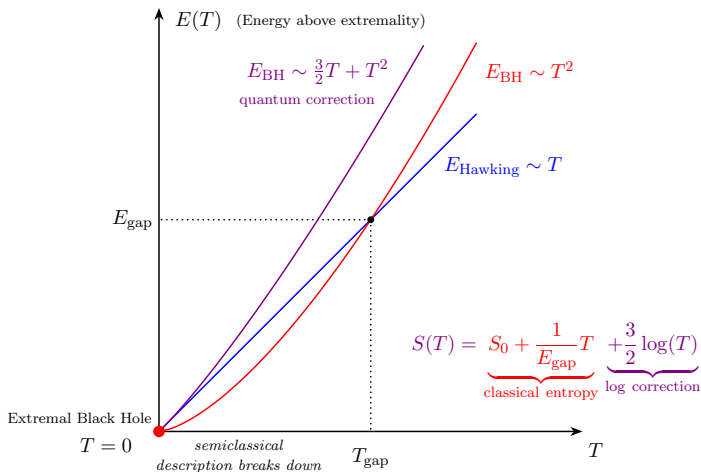
Mass gap puzzle for Reissner–Nordström Black Holes



[Iliesiu, Turiaci; 20]



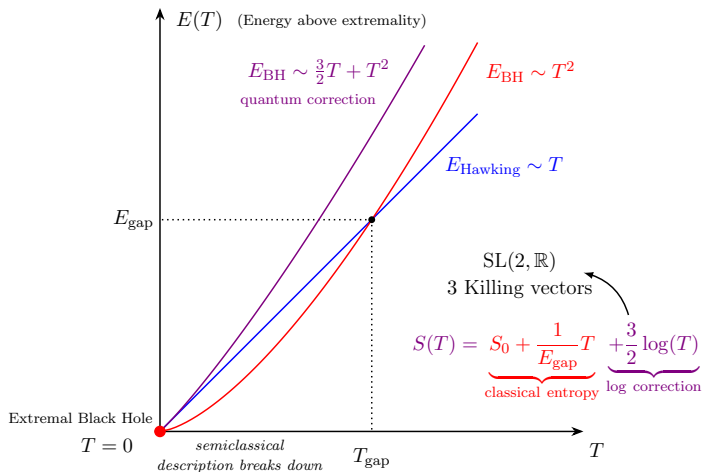
Mass gap puzzle for Reissner–Nordström Black Holes



[Iliesiu, Turiaci; 20]



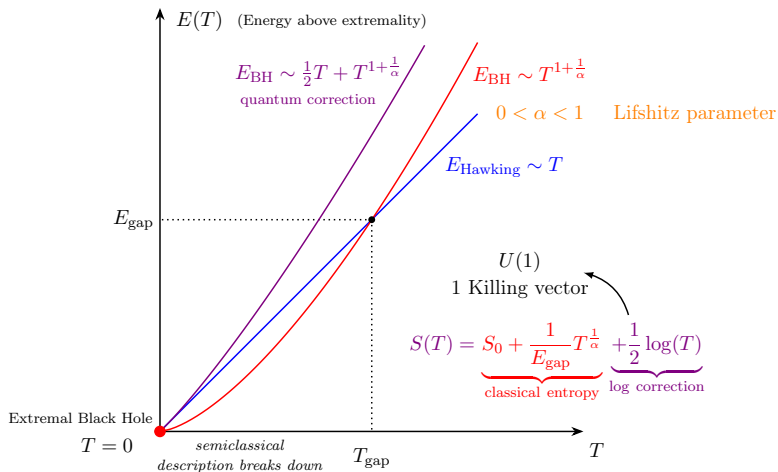
Mass gap puzzle for Reissner–Nordström Black Holes



[Iliesiu, Turiaci; 20]



Mass gap puzzle for Lifshitz Black Holes



[Harsen, Sybesma; 24]



From Near-Extremal Black Holes to 2D Dilaton Models



From Near-Extremal Black Holes to 2D Dilaton Models

Reissner-Nordström



From Near-Extremal Black Holes to 2D Dilaton Models

Reissner-Nordström

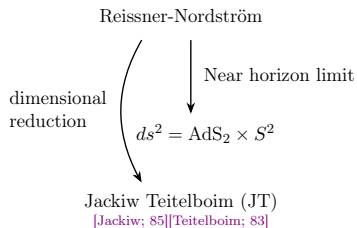


Near horizon limit

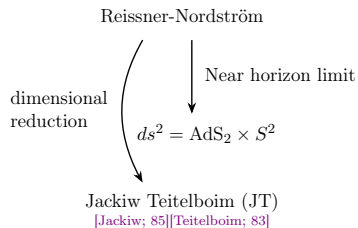
$$ds^2 = \text{AdS}_2 \times S^2$$



From Near-Extremal Black Holes to 2D Dilaton Models



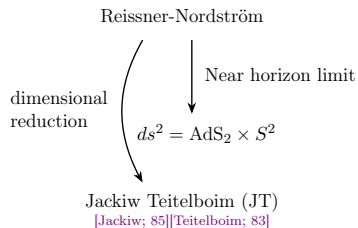
From Near-Extremal Black Holes to 2D Dilaton Models



Lifshitz Black Hole



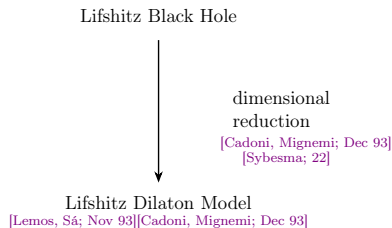
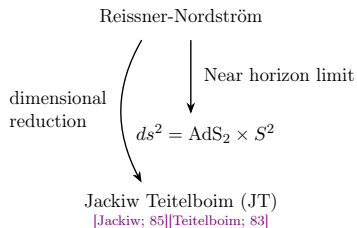
From Near-Extremal Black Holes to 2D Dilaton Models



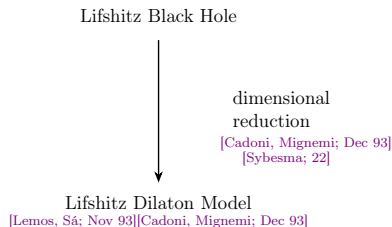
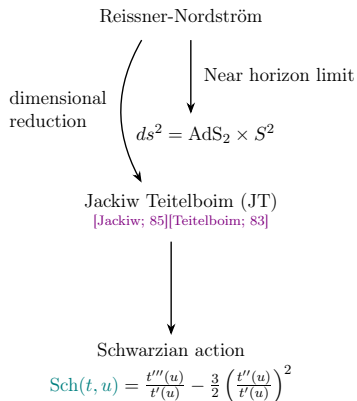
Lifshitz Black Hole



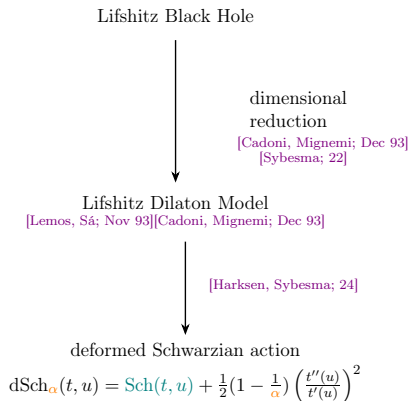
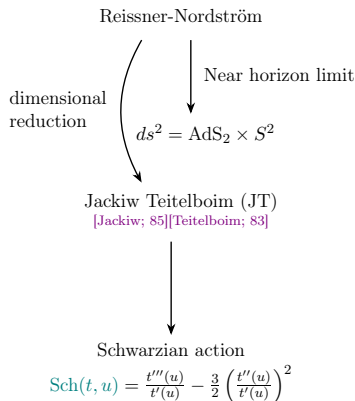
From Near-Extremal Black Holes to 2D Dilaton Models



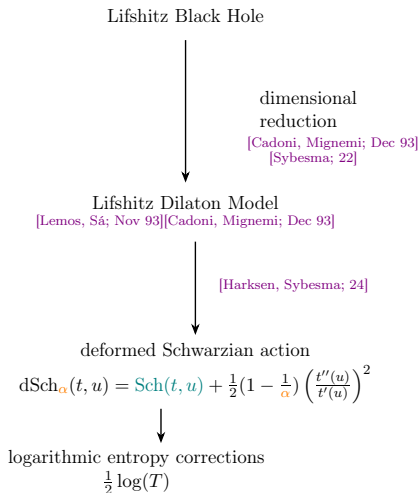
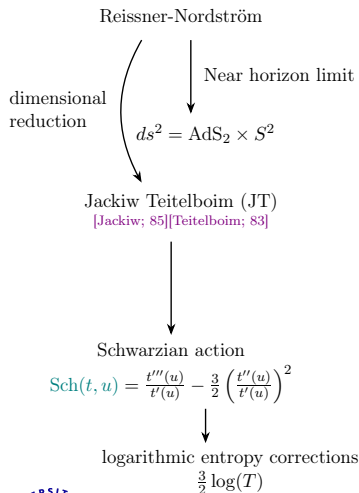
From Near-Extremal Black Holes to 2D Dilaton Models



From Near-Extremal Black Holes to 2D Dilaton Models



From Near-Extremal Black Holes to 2D Dilaton Models



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1 - \alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

$\alpha = 1$: JT gravity [Jackiw; 85][Teitelboim; 83]

$$I_{\text{JT}} = I_{\text{Lif}}(\alpha = 1) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} [R + 4\lambda^2]$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

$\alpha = 1$: JT gravity [Jackiw; 85][Teitelboim; 83]

$$I_{\text{JT}} = I_{\text{Lif}}(\alpha = 1) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} [R + 4\lambda^2]$$

$\alpha = 0$: CGHS gravity [Callan, Giddings, Harvey, Strominger; 92]

$$I_{\text{CGHS}} = I_{\text{Lif}}(\alpha = 0) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} [R + 4(\nabla\phi)^2 + 4\lambda^2]$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Can get rid of **kinetic term**

$$\Phi = e^{-2\phi}, \quad g_{ab} \rightarrow \Phi^{\alpha-1} g_{ab}.$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Can get rid of **kinetic term**

$$\Phi = e^{-2\phi}, \quad g_{ab} \rightarrow \Phi^{\alpha-1} g_{ab}.$$

in which case

$$I_{\text{W-Lif}} = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} (\Phi R + 4\lambda^2 \Phi^\alpha)$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Can get rid of **kinetic term**

$$\Phi = e^{-2\phi}, \quad g_{ab} \rightarrow \Phi^{\alpha-1} g_{ab}.$$

in which case

$$I_{\text{W-Lif}} = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} (\Phi R + 4\lambda^2 \Phi^\alpha)$$

Not asymptotically AdS₂ ☹



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Thermodynamics:



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Thermodynamics: [Cadoni, Mignemi; 93][Kumar, Ray; 95]

$$S \sim T^{\frac{1}{\alpha}}, \quad M = \frac{1}{1+\alpha} TS.$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Thermodynamics: [Cadoni, Mignemi; 93][Kumar, Ray; 95]

$$S \sim T^{\frac{1}{\alpha}}, \quad M = \frac{1}{1+\alpha} TS.$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Thermodynamics: [Cadoni, Mignemi; 93][Kumar, Ray; 95]

$$S \sim T^{\frac{1}{\alpha}}, \quad M = \frac{1}{1+\alpha} TS.$$

Asymptotically Lifshitz Black Hole

$$ds^2 = -r^{2z} f(r) dt^2 + \frac{dr^2}{r^2 f(r)} + r^2 d\vec{x}_{d-1}^2, \quad f(r) = 1 - \left(\frac{r_h}{r}\right)^{d-1+z}.$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Thermodynamics: [Cadoni, Mignemi; 93][Kumar, Ray; 95]

$$S \sim T^{\frac{1}{\alpha}}, \quad M = \frac{1}{1+\alpha} TS.$$

Asymptotically Lifshitz Black Hole

$$ds^2 = -r^{2z} f(r) dt^2 + \frac{dr^2}{r^2 f(r)} + r^2 d\vec{x}_{d-1}^2, \quad f(r) = 1 - \left(\frac{r_h}{r}\right)^{d-1+z}.$$

Thermodynamics: [Taylor; 08][Taylor; 15]

$$S \sim T^{\frac{d-1}{z}}, \quad M = \frac{d-1}{d-1+z} TS.$$



The Lifshitz Dilaton Model

[Lemos, Sá; Nov 93][Cadoni, Mignemi; Dec 93]

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} \underbrace{e^{-2\phi}}_{\text{dilaton}} \left[R + \underbrace{4(1-\alpha)(\nabla\phi)^2}_{\text{kinetic term}} + 4\lambda^2 \right]$$

Thermodynamics: [Cadoni, Mignemi; 93][Kumar, Ray; 95]

$$S \sim T^{\frac{1}{\alpha}}, \quad M = \frac{1}{1+\alpha} TS.$$

Asymptotically Lifshitz Black Hole

$$ds^2 = -r^{2z} f(r) dt^2 + \frac{dr^2}{r^2 f(r)} + r^2 d\vec{x}_{d-1}^2, \quad f(r) = 1 - \left(\frac{r_h}{r}\right)^{d-1+z}.$$

Thermodynamics: [Taylor; 08][Taylor; 15]

[Sybesma; 22]

$$S \sim T^{\frac{d-1}{z}}, \quad M = \frac{d-1}{d-1+z} TS.$$

$$\alpha = \frac{z}{d-1}$$

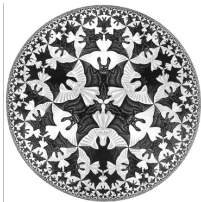


What is the goal?



What is the goal?

[Jackiw; 85][Teitelboim; 83]



JT

[Sachdev, Ye; 93][Kitaev; 15]



What is the goal?

[Jackiw; 85][Teitelboim; 83]



NAdS₂

[Maldacena, Stanford, Yang; 16]

JT

[Sachdev, Ye; 93][Kitaev; 15]

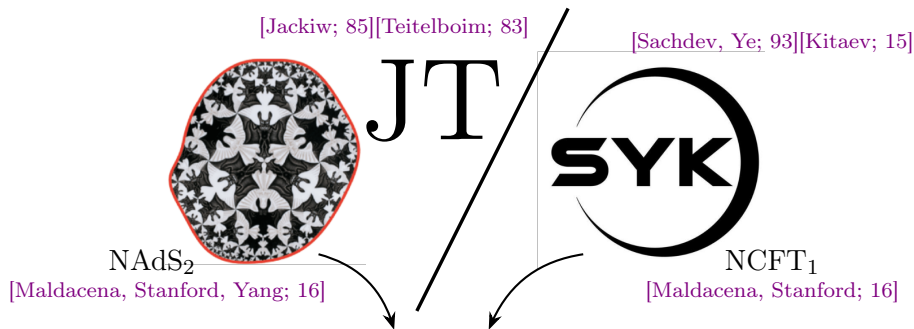


NCFT₁

[Maldacena, Stanford; 16]



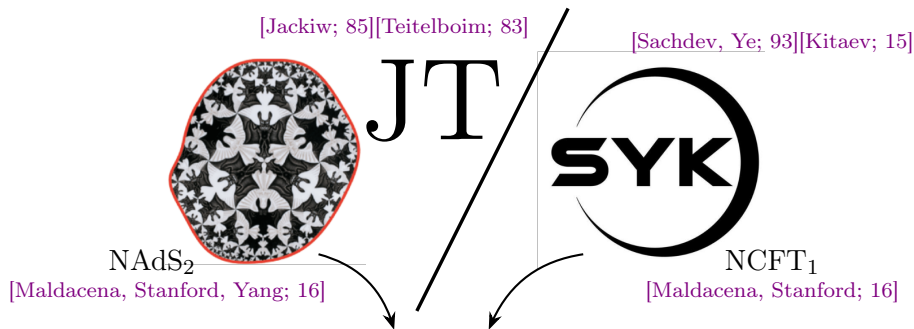
What is the goal?



Schwarzian derivative

$$\text{Sch}(f(z), z) = \frac{f'''(z)}{f''(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)} \right)^2$$

What is the goal?



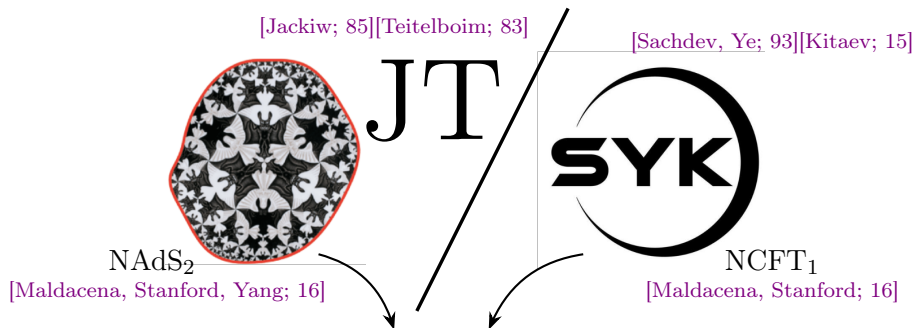
Schwarzian derivative

$$\text{Sch}\left(\frac{Az+B}{Cz+D}, z\right) = 0$$

$$\text{Sch}(f(z), z) = \frac{f'''(z)}{f''(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)} \right)^2$$



What is the goal?



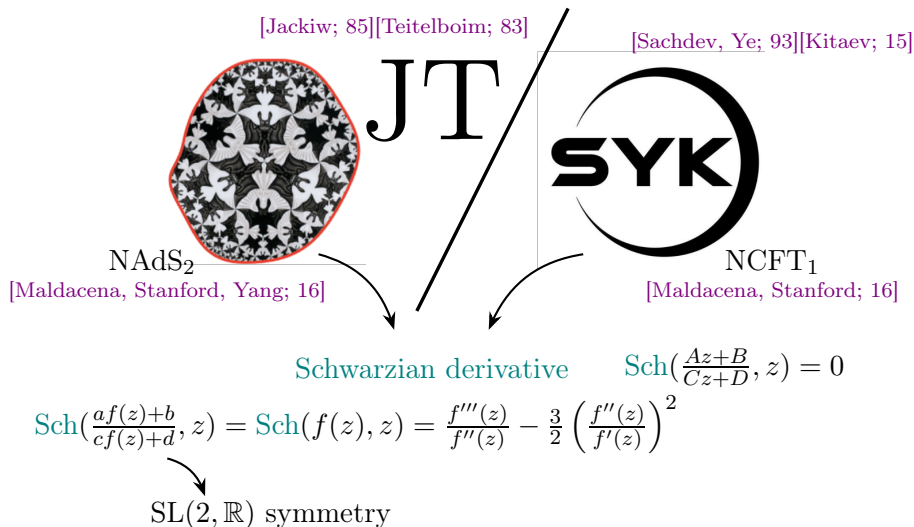
Schwarzian derivative

$$\text{Sch}\left(\frac{Az+B}{Cz+D}, z\right) = 0$$

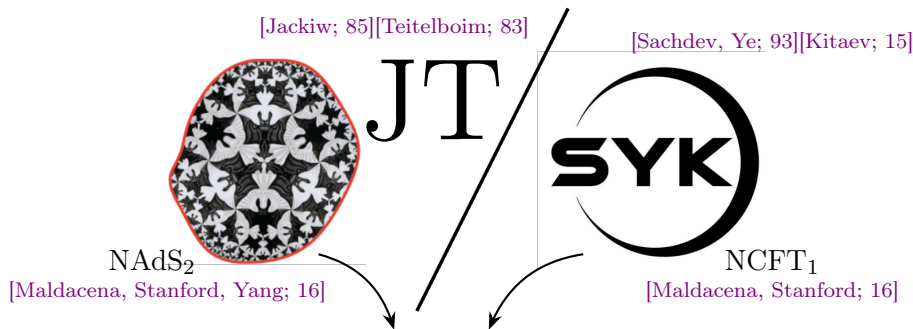
$$\text{Sch}\left(\frac{af(z)+b}{cf(z)+d}, z\right) = \text{Sch}(f(z), z) = \frac{f'''(z)}{f''(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)}\right)^2$$



What is the goal?



What is the goal?



Schwarzian derivative

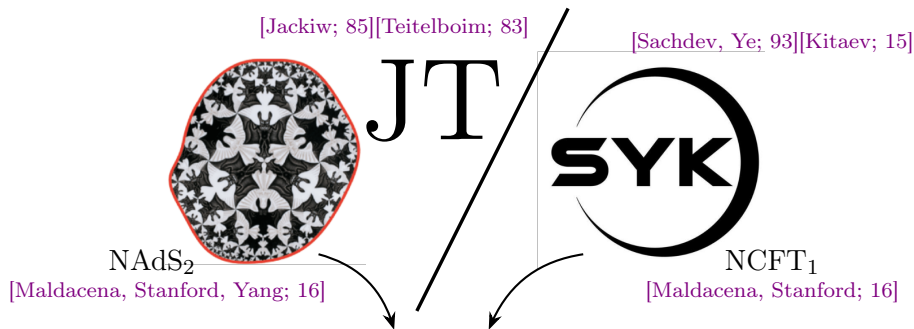
$$\text{Sch}\left(\frac{Az+B}{Cz+D}, z\right) = 0$$

$$\text{Sch}\left(\frac{af(z)+b}{cf(z)+d}, z\right) = \text{Sch}(f(z), z) = \frac{f'''(z)}{f''(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)}\right)^2$$

$SL(2, \mathbb{R})$ symmetry $\xrightarrow{\text{this work}}$ $U(1)$ symmetry



What is the goal?



Schwarzian derivative

$$\text{Sch}\left(\frac{Az+B}{Cz+D}, z\right) = 0$$

$$\text{Sch}\left(\frac{af(z)+b}{cf(z)+d}, z\right) = \text{Sch}(f(z), z) = \frac{f'''(z)}{f''(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)}\right)^2 + \frac{1}{2} \left(1 - \frac{1}{\alpha}\right) \left(\frac{t''(u)}{t'(u)}\right)^2$$

$\text{SL}(2, \mathbb{R})$ symmetry $\xrightarrow{\text{this work}}$ $U(1)$ symmetry



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{Lif}} = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1 + \alpha}} \right)}_{\text{Gibbons-Hawking-York} + \text{counter term}}$$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{Lif}} = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1 + \alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}}$$

EOM:



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{Lif}} = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1 + \alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}}$$

EOM: $\delta\phi :$ $R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{Lif}} = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1 + \alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}}$$

EOM: $\delta\phi :$ $R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} :$ $\nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 \text{EOM} \quad &= - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

EOM: $\delta\phi : \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

EOM: $\delta\phi : \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$



$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$

Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

wiggling boundary



EOM: $\delta\phi :$ $R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} :$ $\nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$

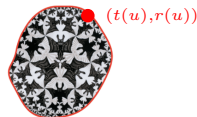
$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}}$ 5/9



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

wiggling boundary



EOM: $\delta\phi : \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$

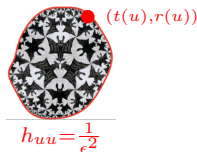
$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

wiggling boundary



EOM: $\delta\phi :$ $R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} :$ $\nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$

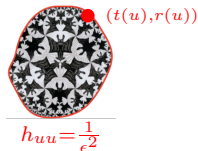
$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}}$ 5/9



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

wiggling boundary



$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

EOM: $\delta\phi : \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$

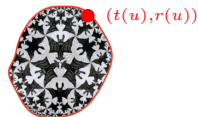
$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

wiggling boundary



$$\sqrt{h} = \frac{1}{\epsilon} + \frac{\epsilon}{2a^2} \left(\frac{t''(u)}{t'(u)} \right)^2 - \frac{b}{2} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{\frac{1}{\alpha}} + \mathcal{O}(\epsilon^3)$$

$$h_{uu} = \frac{1}{\epsilon^2}$$

$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

EOM: $\delta\phi : \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$

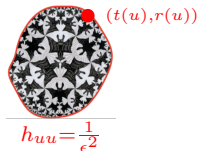
$$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

wiggling boundary



$$\sqrt{h} = \frac{1}{\epsilon} + \frac{\epsilon}{2a^2} \left(\frac{t''(u)}{t'(u)} \right)^2 - \frac{b}{2} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{\frac{1}{\alpha}} + \mathcal{O}(\epsilon^3)$$

$$K = a + \frac{1}{a} \text{Sch}(t, u) \epsilon^2 + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4)$$

$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

EOM: $\delta\phi : \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$$

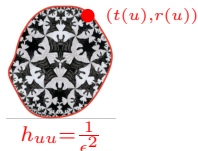


$$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$

Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} \left(R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 \right) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

wiggling boundary



$$\sqrt{h} = \frac{1}{\epsilon} + \frac{\epsilon}{2a^2} \left(\frac{t''(u)}{t'(u)} \right)^2 - \frac{b}{2} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{\frac{1}{\alpha}} + \mathcal{O}(\epsilon^3)$$

$$K = a + \frac{1}{a} \text{Sch}(t, u) \epsilon^2 + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4)$$

$$-2(1 - \alpha) n^\mu \nabla_\mu \phi = \left(\frac{1}{\alpha} - 1 \right) a - \frac{1-\alpha}{2a\alpha} \left(\frac{t''(u)}{t'(u)} \right)^2 \epsilon^2 - \frac{ab}{2} \left(\frac{1}{\alpha} - 1 \right) t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4)$$

$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

EOM: $\delta\phi : \quad R + 4(1 - \alpha) \nabla^2 \phi - 4(1 - \alpha) (\nabla\phi)^2 + 4\lambda^2 = 0$

$$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - \left(\nabla^2 \phi - (1 + \alpha) (\nabla\phi)^2 + \lambda^2 \right) g_{\mu\nu} = 0$$

$$ds^2 = f(r) dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

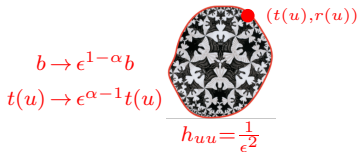
$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} \left(R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 \right) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

$$\sqrt{h} = \frac{1}{\epsilon} + \frac{\epsilon}{2a^2} \left(\frac{t''(u)}{t'(u)} \right)^2 - \frac{b}{2} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{\frac{1}{\alpha}} + \mathcal{O}(\epsilon^3)$$

$$K = a + \frac{1}{a} \text{Sch}(t, u) \epsilon^2 + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4)$$

$$-2(1 - \alpha) n^\mu \nabla_\mu \phi = \left(\frac{1}{\alpha} - 1 \right) a - \frac{1-\alpha}{2a\alpha} \left(\frac{t''(u)}{t'(u)} \right)^2 \epsilon^2 - \frac{ab}{2} \left(\frac{1}{\alpha} - 1 \right) t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4)$$

wiggly boundary



$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

EOM: $\delta\phi: \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

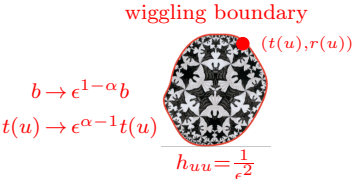
$$\delta g_{\mu\nu}: \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - \left(\nabla^2 \phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2 \right) g_{\mu\nu} = 0$$

$$ds^2 = f(r) dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} \left(R + 4(1-\alpha)(\nabla\phi)^2 + 4\lambda^2 \right) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 \text{EOM} \quad &= - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1-\alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$



$$\sqrt{h} = \frac{1}{\epsilon} + \mathcal{O}(\epsilon^3)$$

$$\begin{aligned}
 K &= a + \frac{1}{a} \text{Sch}(t, u) \epsilon^2 + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4) \\
 -2(1-\alpha) n^\mu \nabla_\mu \phi &= \left(\frac{1}{\alpha} - 1\right) a - \frac{1-\alpha}{2a\alpha} \left(\frac{t''(u)}{t'(u)}\right)^2 \epsilon^2 - \frac{ab}{2} \left(\frac{1}{\alpha} - 1\right) t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4)
 \end{aligned}$$

.....

$$\text{EOM: } \delta\phi : \quad R + 4(1-\alpha)\nabla^2\phi - 4(1-\alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$$



$$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - \left(\nabla^2 \phi - (1+\alpha)(\nabla\phi)^2 + \lambda^2 \right) g_{\mu\nu} = 0$$

$$ds^2 = f(r) dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$

Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

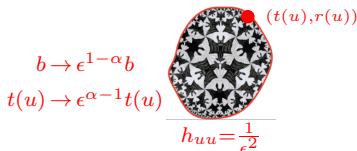
$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 \text{EOM} \quad &= - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

$$\sqrt{h} = \frac{1}{\epsilon} + \mathcal{O}(\epsilon^3)$$

$$K = a + \left(\frac{1}{a} \text{Sch}(t, u) + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \right) \epsilon^2 + \mathcal{O}(\epsilon^4)$$

$$-2(1 - \alpha) n^\mu \nabla_\mu \phi = \left(\frac{1}{\alpha} - 1 \right) a - \frac{1-\alpha}{2a\alpha} \left(\frac{t''(u)}{t'(u)} \right)^2 \epsilon^2 - \frac{ab}{2} \left(\frac{1}{\alpha} - 1 \right) t'(u)^{1+\frac{1}{\alpha}} \epsilon^{1+\frac{1}{\alpha}} + \mathcal{O}(\epsilon^4)$$

wiggly boundary



$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

EOM: $\delta\phi : \quad R + 4(1 - \alpha) \nabla^2 \phi - 4(1 - \alpha) (\nabla\phi)^2 + 4\lambda^2 = 0$

$$\delta g_{\mu\nu} : \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2 \phi - (1 + \alpha) (\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$$

$$ds^2 = f(r) dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

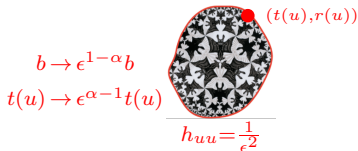
$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 \text{EOM} \quad &= - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)
 \end{aligned}$$

$$\sqrt{h} = \frac{1}{\epsilon} + \mathcal{O}(\epsilon^3)$$

$$K = a + \left(\frac{1}{a} \text{Sch}(t, u) + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \right) \epsilon^2 + \mathcal{O}(\epsilon^4)$$

$$-2(1 - \alpha) n^\mu \nabla_\mu \phi = \left(\frac{1}{\alpha} - 1 \right) a - (1 - \alpha) \left(\frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} + \frac{1}{2a\alpha} \left(\frac{t''(u)}{t'(u)} \right)^2 \right) \epsilon^2 + \mathcal{O}(\epsilon^4)$$

wiggly boundary



$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

EOM: $\delta\phi: \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$$\delta g_{\mu\nu}: \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$$

$$ds^2 = f(r)dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right) \\
 &= - \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \left(\frac{ab}{2} t'(u)^{1+\frac{1}{\alpha}} + \frac{1}{a} \text{Sch}(t, u) + \frac{(1 - \frac{1}{\alpha})}{2a} \left(\frac{t''(u)}{t'(u)} \right)^2 \right) + \mathcal{O}(\epsilon)
 \end{aligned}$$

wiggling boundary



$(t(u), r(u))$

$$b \rightarrow \epsilon^{1-\alpha} b$$

$$t(u) \rightarrow \epsilon^{\alpha-1} t(u)$$

$$h_{uu} = \frac{1}{\epsilon^2}$$

$$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$$

$$\sqrt{h} = \frac{1}{\epsilon} + \mathcal{O}(\epsilon^3)$$

$$K = a + \left(\frac{1}{a} \text{Sch}(t, u) + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \right) \epsilon^2 + \mathcal{O}(\epsilon^4)$$

$$-2(1 - \alpha) n^\mu \nabla_\mu \phi = \left(\frac{1}{\alpha} - 1 \right) a - (1 - \alpha) \left(\frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} + \frac{1}{2a\alpha} \left(\frac{t''(u)}{t'(u)} \right)^2 \right) \epsilon^2 + \mathcal{O}(\epsilon^4)$$

EOM: $\delta\phi: \quad R + 4(1 - \alpha)\nabla^2\phi - 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2 = 0$

$$\delta g_{\mu\nu}: \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2\phi - (1 + \alpha)(\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$$

$$ds^2 = f(r) dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$



Deriving the deformed Schwarzian [Harksen, Sybesma; 24]

$$\begin{aligned}
 I_{\text{Lif}} &= -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} (R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2) - \underbrace{\int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(K - \frac{2\lambda}{\sqrt{1+\alpha}} \right)}_{\text{Gibbons-Hawking-York + counter term}} \\
 &\stackrel{\text{EOM}}{=} - \int_{\partial\mathcal{M}} du \sqrt{h} e^{-2\phi} \left(-2(1 - \alpha) n^\mu \nabla_\mu \phi + K - \frac{2\lambda}{\sqrt{1+\alpha}} \right) \\
 &= - \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \left(\frac{ab}{2} t'(u)^{1+\frac{1}{\alpha}} + \frac{1}{a} \text{Sch}(t, u) + \frac{(1 - \frac{1}{\alpha})}{2a} \left(\frac{t''(u)}{t'(u)} \right)^2 \right) + \mathcal{O}(\epsilon) \\
 &\stackrel{\epsilon \rightarrow 0}{=} \boxed{-\alpha M \int t'(u) du - \frac{1}{a} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} d\text{Sch}_\alpha(t, u)}
 \end{aligned}$$

wiggling boundary



$b \rightarrow \epsilon^{1-\alpha} b$
 $t(u) \rightarrow \epsilon^{\alpha-1} t(u)$

$h_{uu} = \frac{1}{\epsilon^2}$

$$\sqrt{h} = \frac{1}{\epsilon} + \mathcal{O}(\epsilon^3)$$

$$K = a + \left(\frac{1}{a} \text{Sch}(t, u) + \frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} \right) \epsilon^2 + \mathcal{O}(\epsilon^4)$$

$r(u) = \frac{1}{a\epsilon t'(u)} + \mathcal{O}(\epsilon)$

$$-2(1 - \alpha) n^\mu \nabla_\mu \phi = \left(\frac{1}{\alpha} - 1 \right) a - (1 - \alpha) \left(\frac{ab}{2\alpha} t'(u)^{1+\frac{1}{\alpha}} + \frac{1}{2a\alpha} \left(\frac{t''(u)}{t'(u)} \right)^2 \right) \epsilon^2 + \mathcal{O}(\epsilon^4)$$

EOM: $\delta\phi: \quad R + 4(1 - \alpha) \nabla^2 \phi - 4(1 - \alpha) (\nabla\phi)^2 + 4\lambda^2 = 0$

$$\delta g_{\mu\nu}: \quad \nabla_\mu \nabla_\nu \phi - 2\alpha \nabla_\mu \phi \nabla_\nu \phi - (\nabla^2 \phi - (1 + \alpha) (\nabla\phi)^2 + \lambda^2) g_{\mu\nu} = 0$$

$$ds^2 = f(r) dt^2 + \frac{dr^2}{f(r)}, \quad f(r) = (ar)^2 - b(ar)^{1-\frac{1}{\alpha}}, \quad e^{-2\phi} = (ar)^{\frac{1}{\alpha}}, \quad a = \frac{2\alpha\lambda}{\sqrt{1+\alpha}} \quad 5/9$$



Remarks on the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} d\text{Sch}_{\alpha}(t, u)$$



Remarks on the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = \underbrace{-\alpha M \int t'(u) du}_{\text{free energy term}} - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} d\text{Sch}_{\alpha}(t, u)$$



Remarks on the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = \underbrace{-\alpha M \int t'(u) du}_{\text{free energy term}} - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} d\text{Sch}_{\alpha}(t, u)$$

$$\text{where} \quad d\text{Sch}_{\alpha}(t, u) = \text{Sch}(t, u) + \frac{1}{2} \left(1 - \frac{1}{\alpha}\right) \left(\frac{t''(u)}{t'(u)}\right)^2$$



Remarks on the deformed Schwarzian [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = \underbrace{-\alpha M \int t'(u) du}_{\text{free energy term}} - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \text{dSch}_{\alpha}(t, u)$$

$$\text{where} \quad \text{dSch}_{\alpha}(t, u) = \text{Sch}(t, u) + \frac{1}{2} \left(1 - \frac{1}{\alpha}\right) \left(\frac{t''(u)}{t'(u)}\right)^2$$

$$(\text{JT}) \quad \underline{\alpha = 1}: \quad \text{dSch}_{\alpha=1}(t, u) = \text{Sch}(t, u)$$

Classical Partition Function [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} d\text{Sch}_{\alpha}(t, u)$$



Classical Partition Function [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \text{dSch}_{\alpha}(t, u)$$

EOM:

$$\frac{d}{du} \left(\frac{\text{dSch}_{\alpha}}{t'(u)^{1+\frac{1}{\alpha}}} \right) = 0 \quad \implies \quad t(u) = \begin{cases} au + b & \text{if } 0 < \alpha < 1 \\ \frac{au+b}{cu+d} & \text{if } \alpha = 1 \end{cases}$$



Classical Partition Function [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \text{dSch}_{\alpha}(t, u)$$

EOM: $\frac{d}{du} \left(\frac{\text{dSch}_{\alpha}}{t'(u)^{1+\frac{1}{\alpha}}} \right) = 0 \quad \implies \quad t(u) = \begin{cases} au + b & \text{if } 0 < \alpha < 1 \\ \frac{au+b}{cu+d} & \text{if } \alpha = 1 \end{cases}$

On-shell action: $\mathcal{I}_{\text{dSch}} = I_{\text{dSch}}[t = u] = -\alpha M \beta = \beta F \sim \beta^{-\frac{1}{\alpha}}$



Classical Partition Function [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{\alpha} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \text{dSch}_{\alpha}(t, u)$$

EOM: $\frac{d}{du} \left(\frac{\text{dSch}_{\alpha}}{t'(u)^{1+\frac{1}{\alpha}}} \right) = 0 \quad \implies \quad t(u) = \begin{cases} au + b & \text{if } 0 < \alpha < 1 \\ \frac{au+b}{cu+d} & \text{if } \alpha = 1 \end{cases}$

On-shell action: $\mathcal{I}_{\text{dSch}} = I_{\text{dSch}}[t = u] = -\alpha M \beta = \beta F \sim \beta^{-\frac{1}{\alpha}}$

Classical partition function: $Z_{\text{cl}}(\beta) = e^{-\mathcal{I}_{\text{Lif}}} = e^{S_0 + \alpha C \beta^{-\frac{1}{\alpha}}}$



Classical Partition Function [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{a} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \text{dSch}_{\alpha}(t, u)$$

EOM:
$$\frac{d}{du} \left(\frac{\text{dSch}_{\alpha}}{t'(u)^{1+\frac{1}{\alpha}}} \right) = 0 \quad \implies \quad t(u) = \begin{cases} au + b & \text{if } 0 < \alpha < 1 \\ \frac{au+b}{cu+d} & \text{if } \alpha = 1 \end{cases}$$

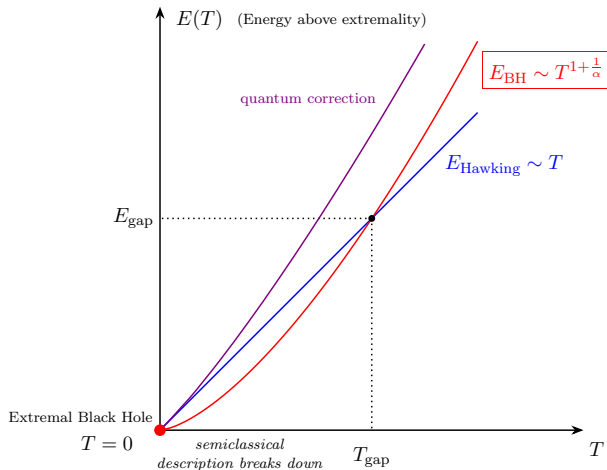
On-shell action:
$$\mathcal{I}_{\text{Lif}} = \mathcal{I}_{\text{dSch}} = I_{\text{dSch}}[t = u] = -\alpha M \beta = \beta F \sim \beta^{-\frac{1}{\alpha}}$$

Classical partition function:
$$Z_{\text{cl}}(\beta) = e^{-\mathcal{I}_{\text{Lif}}} = e^{S_0 + \alpha C \beta^{-\frac{1}{\alpha}}}$$

Energy above extremality:
$$E_{\text{BH}}(\beta) = C \beta^{-(1+\frac{1}{\alpha})}$$



Classical Partition Function [Harksen, Sybesma; 24]



Logarithmic Correction [Harksen, Sybesma; 24]



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)]$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$

$$\epsilon(u) = \frac{\beta}{2\pi} \sum_{n \neq 0} \left(\epsilon_n^{(R)} + i \epsilon_n^{(I)} \right) e^{-\frac{2\pi}{\beta} i n u}$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$

$$\boxed{\epsilon(u) = \frac{\beta}{2\pi} \sum_{n \neq 0} \left(\epsilon_n^{(R)} + i \epsilon_n^{(I)} \right) e^{-\frac{2\pi}{\beta} i n u}} = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{2\pi^2}{\beta \alpha a} \sum_{n \neq 0} n^4 \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$

$$\boxed{\epsilon(u) = \frac{\beta}{2\pi} \sum_{n \neq 0} \left(\epsilon_n^{(R)} + i \epsilon_n^{(I)} \right) e^{-\frac{2\pi}{\beta} i n u}} = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{2\pi^2}{\beta \alpha a} \sum_{n \neq 0} n^4 \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)$$

$$Z_Q(\beta) \approx \int_{\text{Diff}(S^1)/U(1)} [\mathcal{D}t]_{\text{Lif}} e^{-I_{\text{dSch}}[t]}$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$

$$\boxed{\epsilon(u) = \frac{\beta}{2\pi} \sum_{n \neq 0} \left(\epsilon_n^{(R)} + i \epsilon_n^{(I)} \right) e^{-\frac{2\pi}{\beta} i n u}} = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{2\pi^2}{\beta \alpha a} \sum_{n \neq 0} n^4 \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)$$

$$Z_Q(\beta) \approx \int_{\text{Diff}(S^1)/U(1)} [\mathcal{D}t]_{\text{Lif}} e^{-I_{\text{dSch}}[t]}$$

$$\begin{aligned} [\mathcal{D}t]_{\text{JT}} &= \prod_{n \geq 2} 4\pi n(n-1)(n+1) d\epsilon_n^{(R)} d\epsilon_n^{(I)} \\ [\mathcal{D}t]_{\text{Lif}} &= \prod_{n \geq 1} 4\pi n d\epsilon_n^{(R)} d\epsilon_n^{(I)} \end{aligned}$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$

$$\boxed{\epsilon(u) = \frac{\beta}{2\pi} \sum_{n \neq 0} \left(\epsilon_n^{(R)} + i \epsilon_n^{(I)} \right) e^{-\frac{2\pi}{\beta} i n u}} = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{2\pi^2}{\beta \alpha a} \sum_{n \neq 0} n^4 \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)$$

$$Z_Q(\beta) \approx \int_{\text{Diff}(S^1)/U(1)} [\mathcal{D}t]_{\text{Lif}} e^{-I_{\text{dSch}}[t]}$$

$$= e^{\alpha C \beta^{-\frac{1}{\alpha}}} \prod_{n \geq 1} 4\pi n \int d\epsilon_n^{(R)} d\epsilon_n^{(I)} e^{-\frac{2\pi^2 n^4}{\beta \alpha a} \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)}$$

$$\begin{aligned} [\mathcal{D}t]_{\text{JT}} &= \prod_{n \geq 2} 4\pi n (n-1)(n+1) d\epsilon_n^{(R)} d\epsilon_n^{(I)} \\ [\mathcal{D}t]_{\text{Lif}} &= \prod_{n \geq 1} 4\pi n d\epsilon_n^{(R)} d\epsilon_n^{(I)} \end{aligned}$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$

$$\boxed{\epsilon(u) = \frac{\beta}{2\pi} \sum_{n \neq 0} \left(\epsilon_n^{(R)} + i \epsilon_n^{(I)} \right) e^{-\frac{2\pi}{\beta} i n u}} = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{2\pi^2}{\beta \alpha a} \sum_{n \neq 0} n^4 \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)$$

$$Z_Q(\beta) \approx \int_{\text{Diff}(S^1)/U(1)} [\mathcal{D}t]_{\text{Lif}} e^{-I_{\text{dSch}}[t]}$$

$$\begin{aligned} [\mathcal{D}t]_{\text{JT}} &= \prod_{n \geq 2} 4\pi n (n-1)(n+1) d\epsilon_n^{(R)} d\epsilon_n^{(I)} \\ [\mathcal{D}t]_{\text{Lif}} &= \prod_{n \geq 1} 4\pi n d\epsilon_n^{(R)} d\epsilon_n^{(I)} \end{aligned}$$

$$= e^{\alpha C \beta^{-\frac{1}{\alpha}}} \prod_{n \geq 1} 4\pi n \int d\epsilon_n^{(R)} d\epsilon_n^{(I)} e^{-\frac{2\pi^2 n^4}{\beta \alpha a} \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)}$$

$$= \frac{1}{4\pi^{3/2} \sqrt{a\alpha\beta}} e^{\alpha C \beta^{-\frac{1}{\alpha}}} \implies S(T) \sim S_0 + \frac{1}{E_{\text{gap}}} T^{\frac{1}{\alpha}} + \frac{1}{2} \log(T)$$



Logarithmic Correction [Harksen, Sybesma; 24]

$$I_{\text{dSch}}[t = u + \epsilon(u)] = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{1}{2a\alpha} \int_0^\beta du \epsilon''(u)^2 + \mathcal{O}(\epsilon^3)$$

$$\boxed{\epsilon(u) = \frac{\beta}{2\pi} \sum_{n \neq 0} \left(\epsilon_n^{(R)} + i \epsilon_n^{(I)} \right) e^{-\frac{2\pi}{\beta} i n u}} = -\alpha C \beta^{-\frac{1}{\alpha}} + \frac{2\pi^2}{\beta \alpha a} \sum_{n \neq 0} n^4 \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)$$

$$Z_Q(\beta) \approx \int_{\text{Diff}(S^1)/U(1)} [\mathcal{D}t]_{\text{Lif}} e^{-I_{\text{dSch}}[t]}$$

$$= e^{\alpha C \beta^{-\frac{1}{\alpha}}} \prod_{n \geq 1} 4\pi n \int d\epsilon_n^{(R)} d\epsilon_n^{(I)} e^{-\frac{2\pi^2 n^4}{\beta \alpha a} \left(\epsilon_n^{(R)} \epsilon_n^{(R)} + \epsilon_n^{(I)} \epsilon_n^{(I)} \right)}$$

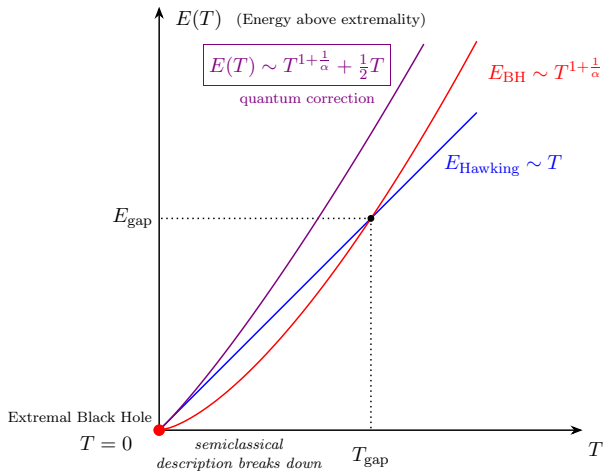
$$= \frac{1}{4\pi^{3/2} \sqrt{a\alpha\beta}} e^{\alpha C \beta^{-\frac{1}{\alpha}}}$$

$$\Rightarrow S(T) \sim S_0 + \frac{1}{E_{\text{gap}}} T^{\frac{1}{\alpha}} + \frac{1}{2} \log(T)$$

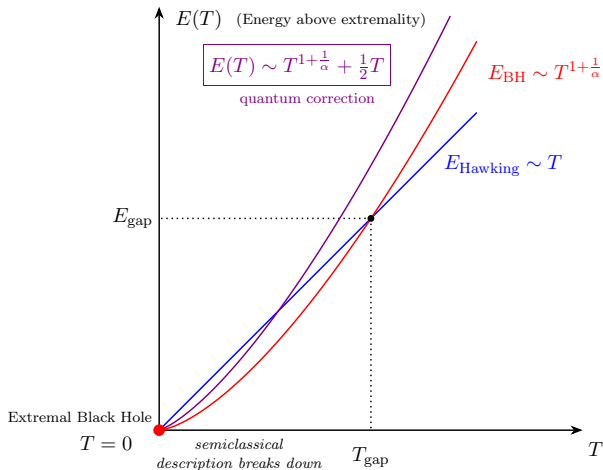
$$E(T) \sim T^{1+\frac{1}{\alpha}} + \frac{1}{2} T$$



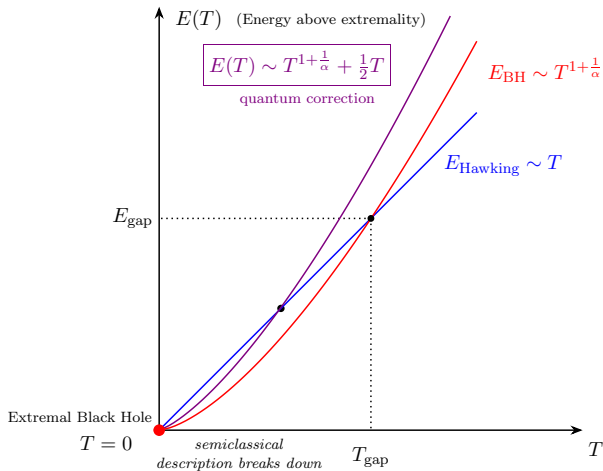
Logarithmic Correction [Harksen, Sybesma; 24]



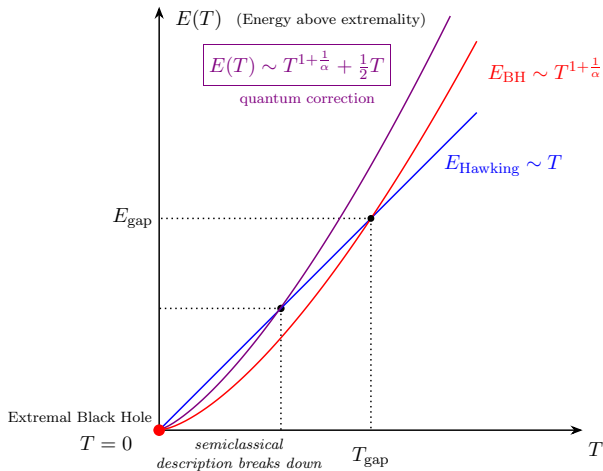
Logarithmic Correction [Harsen, Sybesma; 24]



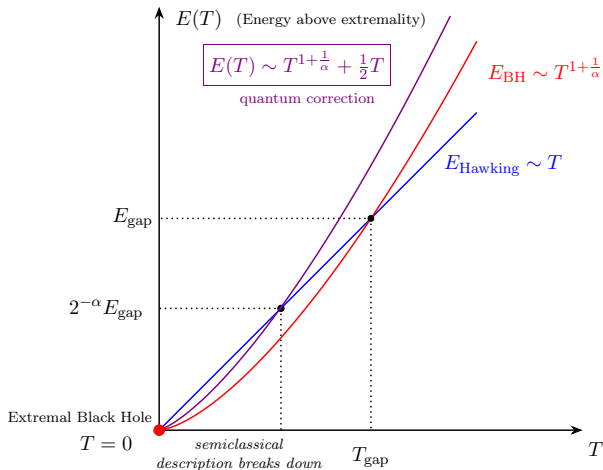
Logarithmic Correction [Harsen, Sybesma; 24]



Logarithmic Correction [Harksen, Sybesma; 24]



Logarithmic Correction [Harsen, Sybesma; 24]



Summary and Conclusion



Summary and Conclusion

□ Examined the $2D$ dilaton model:

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} [R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2] .$$

which arises in the dimensional reduction of a Lifshitz black hole.



Summary and Conclusion

□ Examined the $2D$ dilaton model:

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} [R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2] .$$

which arises in the dimensional reduction of a Lifshitz black hole.

△ Derived a deformed Schwarzian action

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{a} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \text{dSch}_{\alpha}(t, u) .$$

which breaks the $\text{SL}(2, \mathbb{R})$ symmetry down to $U(1)$.



Summary and Conclusion

□ Examined the $2D$ dilaton model:

$$I_{\text{Lif}}(\alpha) = -\frac{1}{2} \int_{\mathcal{M}} d^2x \sqrt{g} e^{-2\phi} [R + 4(1 - \alpha)(\nabla\phi)^2 + 4\lambda^2] .$$

which arises in the dimensional reduction of a Lifshitz black hole.

△ Derived a deformed Schwarzian action

$$I_{\text{dSch}}[t] = -\alpha M \int t'(u) du - \frac{1}{a} \int du \frac{1}{t'(u)^{\frac{1}{\alpha}}} \text{dSch}_{\alpha}(t, u) .$$

which breaks the $\text{SL}(2, \mathbb{R})$ symmetry down to $U(1)$.

○ Computed a logarithmic correction to the entropy

$$S(T) = S_0 + \frac{1}{E_{\text{gap}}} T^{\frac{1}{\alpha}} + \frac{1}{2} \log(T) .$$

and lowers the mass-gap scale.

